Sofia Lindelow

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Forward Deployed AI Engineer and psychology student at McGill, building AI applications and conducting research across health psychology, computational neuroscience and molecular modeling. Builds agent workflows with tests, tracing, evals and human approval.

EDUCATION

McGill University | BSc Psychology, Minor in Interdisciplinary Life Sciences | Montreal, QC | *Expected June 2027*

- Cumulative GPA: 3.68 / 4.0. Coursework includes Integrative Neuroscience, Behavioral Neuroscience, Cognition, Psychological Statistics, Statistics for Experimental Design, Mammalian Physiology I and II, Human Perception Laboratory (MATLAB), Foundations of Programming (Python), Global Health, and Psychology Research Experience. Planned for 2026 to 2027: Circuitry of the Human Brain, Human Decision-Making, Sensorimotor Neuroscience, Human Memory, and Hormones and Behavior.

University of New South Wales | Semester Abroad | Sydney, Australia | *2026*

- Cognitive Science: 81, Distinction; Social and Developmental Psychology: 81, Distinction; Creative Entrepreneurship: 88, High Distinction; Inside the Criminal Mind: 76, Distinction.

PROFESSIONAL EXPERIENCE

Forward Deployed AI Engineer | Lake House Group | Montreal, QC | *July 2026 to Present*

- Own assigned AI delivery from workflow discovery and scoping through implementation, testing and user handoff.
- Developed Python workflows that translate approved designs into native email drafts through REST APIs, with asset processing, structured manifests, destination-link checks, account validation and saved-output readback; validated draft creation through user testing.
- Built and internally validated a Python reporting pipeline that reconciles source data, calculates metrics deterministically and prepares evidence-backed analysis for human review, with schema, reporting-period and completeness checks so figures stay traceable to their inputs.
- Developed TypeScript agent and workflow components with Mastra and PostgreSQL-backed execution state: structured tool contracts, scoped authorization, approval boundaries and recovery checks, including tests for interrupted processes and database restarts, within a pre-release agent runtime.
- Implemented Langfuse tracing and evaluation workflows to inspect model calls, timing, token usage, costs and output quality; verified development experiments through persisted score readback.
- Built reproducible verification workflows using Python test harnesses, GitHub Actions and Playwright browser checks, and packaged installation, troubleshooting and operator documentation with each implementation.
- Co-lead internal AI enablement through workshops, practical exercises and reusable skills and plugins, helping colleagues set up development environments, use agent workflows and take over delivered tools.

Research Assistant | McGill Health Psychology Lab | Montreal, QC | *April 2025 to Present*

- Refine participant instructions and survey measures and test cursor-tracking procedures for a study of uncertainty in self-reported readiness for behavior change (Qualtrics, Prolific).
- Contributed a four-week intervention protocol, selection of validated questionnaires and construct-to-question mapping; supported sampling, preregistration and ethics preparation.
- Co-lead an umbrella-review protocol on how systematic reviews and meta-analyses report stage-of-change measurement and its relationship to intervention outcomes; develop the search, appraisal and extraction methods.

AI Training Expert | AfterQuery | Remote | *November 2025 to February 2026*

- Produced expert training examples for AI models across scientific research, education and finance, and refined evaluation rubrics against domain-specific quality criteria.
- Assessed model self-grading and output equivalence across evaluation frameworks, reviewing whether scoring methods captured the quality of generated responses.

Staff Writer, Science and Technology | The Tribune | Montreal, QC | *September 2025 to Present*

- Interview researchers and translate findings in neuroscience, physiology, psychology and global health into published research briefs for a general readership, explaining methods, findings and limitations.

Growth and Development Intern | BetaTown Venture Studio | New York, NY | *May to August 2025*

- Built an internal CRM to organize creator and domain-expert outreach, track contacts and support user discovery for early-stage AI media ventures.
- Led outreach and market research on AI-assisted publishing, content verification and creator workflows; turned findings into product-strategy work, investor materials and pitch decks.

Marketing and Strategy Intern | Doccla | London, UK | *May to August 2024*

- Used Google Analytics and traffic analysis to guide SEO, campaigns and content for a virtual-ward platform.
- Produced patient-story videos, newsletters, web copy and partner outreach for geriatric care, COPD and diabetes programs.

Research Intern | National Institutes of Health | Washington, DC | *January to August 2023*

- Collected participant data and kept consent records for clinical research on chronic back pain in active-duty pilots.

AI SYSTEMS AND RESEARCH PROJECTS

RELAY, Slack-native customer-success agent | Code · Demo

- Awarded Second Place, New Slack Agent category, 2026 Slack Agent Builder Challenge. Detects unanswered questions in Slack Connect channels, tracks response deadlines and prepares cited replies for human approval; validated the core loop end to end in a real Slack Connect workspace.
- FastAPI backend with Slack OAuth and interactive review controls, Celery/Redis workers, and PostgreSQL/pgvector for question state and semantic retrieval. Connected HubSpot, GitHub and permission-aware Slack search; added an MCP context server, row-level security and encrypted connector tokens.

EEG and Subsequent-Memory Prediction, independent research | Code and report

- Asked whether study-period EEG adds information about later recall beyond behavioral context, and directed AI-assisted development of a reproducible pipeline on real PEERS data: item-level recall labels, EEG preprocessing and epoching, 128 spectral features, and behavioral, EEG-only and combined logistic-regression models.
- Required train-only normalization, session-out predictions for the combined model and a held-session evaluation per participant; locked trial and split manifests and bound outputs to provenance records. Across 38 participants, EEG showed no detectable incremental value over the behavioral baseline (combined minus behavioral AUC -0.0044, 95% CI -0.0121 to 0.0033).

Kinase-Ligand Binding-Affinity Prediction | Code and results

- Led AI-assisted development of an affinity-prediction and retrospective screening tool on DAVIS and KIBA, with frozen source data, saved splits and metrics generated from code. Compared RDKit fingerprint baselines with models adding amino-acid/dipeptide composition or frozen ESM-2 embeddings, using gradient-boosted models and a Chemprop message-passing neural network.
- Tested scaffold and protein holdouts, then MMseqs2 sequence-cluster holdouts on DAVIS. An apparent protein-feature gain on held-out kinases disappeared at 40% sequence identity (Spearman difference -0.0024, 95% CI -0.0277 to 0.0236), a pattern consistent with homology leakage.

Crystal Graph Neural Network, guided CaMML modeling project

- Implemented a crystal-property prediction model in PyTorch with atom and bond features as graph inputs, gated message-passing layers, residual updates, crystal-level mean pooling and a regression head; wrote batched training and evaluation loops with loss and MAE tracking and completed a 40-epoch CPU training run.
- Ran separate validation sweeps on convolution depth and embedding size (favoring four layers and 128-dimensional embeddings, respectively) and kept the findings separate because the combined configuration was not tested.

HIPAA-Guard, healthcare code-review package | Code · PyPI

- Built a multi-agent code-review pipeline combining deterministic Python regex and AST checks with AI analysis, proposed fixes, adversarial review and rescan verification; added redaction before model analysis and shipped it as a CLI with pre-commit integration, published on PyPI.

Earlier agent prototypes

- **Sales development workflows** (inbound, outbound): Python pipelines for inbox ingestion, lead enrichment, urgency scoring and reply classification, keeping model-generated language separate from deterministic routing.
- **Cart intent classifier** (code): Anthropic-powered classifier turning purchase, engagement and browsing signals into structured intent and proposed recovery-email flows, with FastAPI webhooks, Redis and Shopify/Klaviyo API clients.
- **Growth and referral agent** (code) and **competitive intelligence agent** (code): deterministic lead scoring with AI-written messaging, and source-backed competitor research reports in HTML and Google Sheets.

TEACHING AND CAMPUS LEADERSHIP

Pilates, Barre and Yoga Instructor | LeanBean Fitness and independent (Sydney, 2026); MVMT Pilates (Montreal, 2025); CorePower Yoga (Washington, DC, 2023)

- Designed and delivered group classes and individual sessions, adapting progressions and cueing to different experience levels. Recognized as CorePower Yoga Teacher of the Month during my first month at the studio.
- **Campus:** Engagement and Media Coordinator, BABES McGill (February 2025 to Present); Marketing and Media Coordinator, McGill Nourish (January 2025 to Present).

SKILLS, CERTIFICATIONS AND WRITING

AI and Engineering: Python, TypeScript, REST APIs, FastAPI, Mastra, PostgreSQL/pgvector, SQLAlchemy, Celery/Redis, Slack Bolt, OAuth, MCP, Anthropic API, OpenRouter, Langfuse tracing and evals, Playwright, GitHub Actions, Git/GitHub, Render, Supabase, HTML/CSS/JavaScript, agent and workflow design, human-in-the-loop systems.

Machine Learning and Data: PyTorch, scikit-learn, logistic regression, gradient boosting, graph neural networks, Chemprop, ESM-2, RDKit, MMseqs2, MNE, NumPy, pandas, SciPy, leakage-safe evaluation design, bootstrap CIs, pytest.

Research and Delivery: Workflow discovery, technical scoping, user testing, operator documentation, experimental design, behavioral study design, Qualtrics, Prolific, cursor tracking, literature review, statistics, MATLAB, scientific writing.

Languages: English (fluent), Swedish (proficient), Spanish (proficient), Portuguese (basic).

Certifications: Anthropic, July 2026: Claude Code 101, AI Fluency for Students, Model Context Protocol: Advanced Topics. CorePower Yoga 200-hour Yoga Teacher Certification and 50-hour Sculpt Certification; Pilates instructor.

Selected Writing: How to Set Up an Agentic Workspace for Ecommerce (Lake House Group, September 2026); science briefs for The Tribune; Substack.